

# **Trial Title: Digital Diabetes Remission Trial (DIGEST)**

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| <b>Version</b>             | 2.2                                                            |
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## Chief Investigator Agreement for the Protocol

I agree to;

- To assume responsibility for the proper conduct of the clinical investigation at this site, and to conduct the trial in compliance with this protocol, any future amendments, and with any other trial's conduct procedures provided by the sponsor or authorised representatives.
- Not to implement any deviations from or changes to the protocol without agreement from the sponsor and prior review and favourable opinion from the ethics committee and approval from the competent authority, if applicable, except where necessary to eliminate an immediate hazard to the subject(s), or for administrative aspects of the clinical investigation (where permitted by all applicable regulatory requirements).
- To ensure that all persons assisting me with the clinical investigations are adequately informed about the protocol and of their trial-related duties and functions.

Professor Carel le Roux

Chief Investigator Name

.....

.....

Chief Investigator Signature

Date

The signatures below constitute the approval of this protocol and the attachments

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## 1 ABBREVIATIONS

|           |                                           |
|-----------|-------------------------------------------|
| ADA       | American Diabetic Association             |
| AE        | Adverse Events                            |
| AESI      | AEs of Special Interest                   |
| App       | Application                               |
| BMI       | Body Mass Index                           |
| BP        | Blood Pressure                            |
| CI        | Chief Investigator                        |
| e-Consent | Electronic consent                        |
| eTMF      | Electronic Trial Master File              |
| EDC       | Electronic Data Capture                   |
| FAS       | Full Analysis Set                         |
| GCP       | Good Clinical Practice                    |
| GP        | General Practitioner                      |
| HbA1c     | Glycated haemoglobin                      |
| ICF       | Informed Consent Form                     |
| ICH       | International Conference on Harmonisation |
| ITI       | Intention to treat                        |
| LH        | Lindus Health                             |
| MACE      | Major Adverse Cardiovascular Events       |
| MADE      | Major Adverse Diabetes Events             |
| PIS       | Patient Information Sheet                 |
| PPS       | Per Protocol Analysis Set                 |
| REC       | Research Ethics Committee                 |



|     |                           |
|-----|---------------------------|
| RCT | Randomised Clinical Trial |
| RN  | Research Nurse            |
| SAE | Serious Adverse Event     |
| SAP | Statistical Analysis Plan |
| T2D | Type 2 diabetes mellitus  |
| TDR | Total Diet Replacement    |
| TSC | Trial Steering Committee  |

**KEYWORDS:** Diabetic, digital therapeutics, remission, total diet replacement (TDR).

## 2 STUDY SYNOPSIS

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title</b>                | DIGEST: Digital diabetes remission trial                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Design</b>               | An open-label, prospective, parallel design, randomised study assessing Habitual's Remission Programme (total diet replacement and digital therapeutic) versus standard care in achieving remission in type 2 diabetes mellitus (T2D).                                                                                                                                                                                                |
| <b>Phase</b>                | IV                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Recruitment Target</b>   | 100                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Duration</b>             | Main Randomised Controlled Trial (RCT) trial: 6 months<br>Observational period: 12 months                                                                                                                                                                                                                                                                                                                                             |
| <b>Background</b>           | Diabetes remission programmes have recently shown consistently reliable results in patients with T2D, improving many areas of diabetes control, glycaemia, blood pressure, lipids and microvascular damage. However, such programmes may require intensive face to face visits with Health Care Professionals, resulting in increased costs and low uptake and compliance by patients.                                                |
| <b>Aims and Objectives</b>  | The aim of this study is to assess whether diet replacement and a digital therapeutic (Habitual Remission Programme) delivered remotely, leads to remission and weight loss in T2D patients, providing a more cost and resource effective TDR remission programme.<br><br><i>Remission is defined as HbA1c of less than 6.5% (&lt;48 mmol/mol) after at least 2 months off all glucose-lowering medication, measured at 6 months.</i> |
| <b>Recruitment Strategy</b> | GP sites will identify potential participants, who will then be recruited centrally by the Lindus Health research team.                                                                                                                                                                                                                                                                                                               |
| <b>Follow-up</b>            | The study will be conducted remotely, participants will enter study data directly into the Lindus Health Electronic Data Collection (EDC) platform. Participants will collect blood HbA1c samples at home, which will be sent for central processing.                                                                                                                                                                                 |
| <b>Study Population</b>     | Participants with T2D diagnosed within the previous 6 years who are not treated with insulin or GLP-1 RAs.                                                                                                                                                                                                                                                                                                                            |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>Intervention</b>         | <p>Habitual Remission Programme: 3 months of TDR (providing 800 kcal/day) followed by 6 weeks of diet reintroduction, followed by another 6 weeks of further nutritional education &amp; management of caloric intake. This will be supported by the Habitual digital therapeutic, delivered via mobile application.</p> <p>For participants in the intervention arm all glucose-lowering medications will be stopped prior to starting TDR.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Comparator Group</b>     | Standard usual care                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Randomisation Groups</b> | <p>Participants will be randomised in a 2:1 ratio:</p> <ul style="list-style-type: none"> <li>Control arm: standard usual care</li> <li>Intervention arm: Habitual Remission Programme</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Primary Outcomes</b>     | <p><b>Main RCT</b></p> <p>No of participants with:</p> <ul style="list-style-type: none"> <li>Weight loss of 15 kg or more at 6 months<br/>or</li> <li>HbA1c of less than 6.5% (&lt;48 mmol/mol) after at least 2 months off all glucose-lowering medication, measured at 6 months</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Secondary Outcomes</b>   | <p><b>Main RCT</b></p> <ol style="list-style-type: none"> <li>Changes in the following from Baseline to 3 and 6 months: <ul style="list-style-type: none"> <li>Glycaemic control</li> <li>Weight</li> <li>Waist circumference</li> <li>Systolic blood pressure</li> <li>Diastolic blood pressure</li> <li>Medication use</li> </ul> </li> <li>No of participants with the following at 6 months: <ul style="list-style-type: none"> <li>Weight loss of 10 kg or more<br/>or</li> <li>HbA1c of less than 6.5% (&lt;48 mmol/mol) after at least 2 months off all glucose-lowering medication</li> </ul> </li> <li>Intervention adherence - Number of participants starting the intervention at baseline compared to those completing the intervention at 3 and 6 months.</li> <li>Safety: <ul style="list-style-type: none"> <li>Number of AEs</li> <li>Number of SAEs</li> <li>Number of (S)AEs that constitute Major Adverse Cardiovascular Events and Major Adverse Diabetes Events</li> </ul> </li> </ol> <p><b>Observational study (intervention group only)</b></p> |

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <ol style="list-style-type: none"> <li>1. No of participants at 12 months with: <ul style="list-style-type: none"> <li>● Weight loss of 15 kg or more<br/>or</li> <li>● HbA1c of less than 6.5% (&lt;48 mmol/mol) after at least 2 months off all glucose-lowering medication<br/>and</li> </ul> </li> <li>2. No of participants at 12 months with: <ul style="list-style-type: none"> <li>● Weight loss of 10 kg or more<br/>or</li> <li>● HbA1c of less than 6.5% (&lt;48 mmol/mol) after at least 2 months off all glucose-lowering medication<br/>and</li> </ul> </li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Inclusion Criteria</b> | <p>Participants must meet all of the following criteria:</p> <ol style="list-style-type: none"> <li>1. Able and willing to give consent for the study prior to participation</li> <li>2. Be aged 18 - 75 years, with type 2 diabetes mellitus of duration &lt;6 years</li> <li>3. Have access to a smartphone or a computer</li> <li>4. Have a Body Mass Index (BMI) of at least 28 kg/m<sup>2</sup></li> <li>5. HbA1c &gt;48 mmol/mol (6.5%) and ≤86 mmol/mol (10%), within the previous 12 months</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Exclusion Criteria</b> | <p>Participants will be excluded if they meet any of the following criteria:</p> <ol style="list-style-type: none"> <li>1. Are currently using insulin</li> <li>2. Weight change of &gt;5% in the past 3 months</li> <li>3. Have a history or are known to be suffering with alcohol/substance abuse</li> <li>4. Have cancer or currently under investigation for cancer</li> <li>5. Have had a myocardial infarction within the previous 6 months</li> <li>6. Have severe or unstable heart failure e.g. NYHA grade IV</li> <li>7. Have porphyria</li> <li>8. Have learning difficulties</li> <li>9. Are currently on treatment with anti-obesity drugs</li> <li>10. Have had bariatric surgery</li> <li>11. Have been diagnosed with eating disorder or purging</li> <li>12. Are pregnant or less than 4 months postpartum or considering pregnancy in the next 2 years</li> <li>13. Are currently breastfeeding</li> <li>14. Have required hospitalisation for depression or taking antipsychotic drug</li> <li>15. Have a history of illnesses that could interfere with the interpretation of the study results (e.g. HIV, Cushing syndrome, chronic kidney disease, chronic liver disease, hyperthyroidism, hereditary fructose intolerance, depression or antipsychotic drug use within the past 2 years)</li> <li>16. Currently taking Glucagon-like peptide-1 receptor agonists (GLP-1 RAs)</li> <li>17. Have pancreatitis</li> </ol> |

- |  |                                                                                                                                                                                                         |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 18. Currently taking part in a CTIMP trial for antidiabetic medication<br>19. Abnormal diabetic foot review (QOF codes for diabetic foot at moderate risk, at high risk, at increased risk, ulcerated). |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3 BACKGROUND and RATIONALE

#### 3.1 Background

Diabetes increases rates of mortality and morbidity as well as quality of life and presents a major and increasing economic burden currently accounting for 10% of total healthcare expenditure in the UK (1) and 12.5% in the USA (2). More than 4.9 million people in the UK have diabetes, where type 2 diabetes accounts for approximately 90% of this figure (3). A further 13.6 million people are now at increased risk of type 2 diabetes (3). T2D was previously considered a permanent and progressive chronic disease, until more recent evidence demonstrated that remission of type 2 diabetes is possible (4,5). Diabetes UK and American Diabetic Association defines remission as a sustained metabolic improvement where glycated haemoglobin (HbA<sub>1c</sub>) levels return to below 6.5% (48mmol/mol) and which is sustained for at least 3 months in the absence of glucose-lowering medications (12,13).

In the last five years, trials such as DiRECT and DIADEM-I have shown that modest weight loss of approximately 5-10% using diet and lifestyle approaches, can improve all areas of diabetes control including glycaemia, blood pressure, lipids, quality of life and fewer comorbidity complications (6,7). In the Diabetes Remission Clinical Trial (DiRECT), an integrated diet programme delivered entirely within primary care produced remissions of type 2 diabetes (non-diabetic HbA<sub>1c</sub> on no glucose-lowering medication) in 46% of participants at 1 year and 24% of these participants achieved a target weight loss of 15kg or more. T2D remission was maintained in 36% of participants at 2 years (6,8). In addition, the trial showed that participants who underwent a total diet replacement (TDR) and behaviour change programme experienced improved quality of life, decreased need for GP appointments, showed fewer nutritional deficiencies, and fewer medications (6). DIADEM-I, conducted in the Middle East, achieved similar results to DiRECT with 61% of the participants on the intervention arm achieving remission at 12 months, and average weight loss of 12kg (7).

#### 3.2 Rationale

The cost of direct care for patients with type 2 diabetes mellitus (T2D) in the UK is expected to rise from £8.8 billion to £13 billion by 2035/6 (9,10), highlighting urgent need for strategies to achieve T2D remission.

The National Diabetes audit 2019-2020 (11) shows that although 88% of people diagnosed with T2D and other diabetes were offered structured education, there was only 16% attendance. The DIGEST study will examine the effectiveness of a remotely delivered TDR and digital therapeutic intervention to achieve T2D remission, which may be more cost effective to deliver with greater accessibility for patients, and therefore subsequently lead to substantial long-term health gains and cost savings.

The Habitual Remission Programme, consists of 3 months of diet replacements, followed-up by a further 3 months of diet reintroduction (6 weeks of food reintroduction, followed by 6 weeks of nutrition management), combined with a digital behaviour change/tracking/support programme and App.

The main aim of this study is to assess whether diet replacement and a digital therapeutic (Habitual Remission Programme) is more likely to lead to remission by substantial weight loss, compared to standard usual care in adults with T2D diagnosed within the last 6 years. Remission is defined as HbA1c of less than 6.5% (<48 mmol/mol) after at least 2 months off all glucose-lowering medication, measured at 6 months.

## 4 OBJECTIVES AND OUTCOME MEASURES

### 4.1 Randomised Controlled Trial

| Primary Objective                                                                                                                                                                                            | Outcome Measures                                                                                                                                                                               | Timepoint(s) of evaluation of this outcome measure                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>To assess whether diet replacement and a digital therapeutic (Habitual Remission Programme) leads to:</p> <p>i) Weight loss of at least 15kg in T2D patients<br/>or<br/>ii) Remission in T2D patients</p> | <p>Number of participants with:</p> <p>i) Weight loss of 15kg or more<br/>or<br/>ii) HbA1c of less than 6.5% (&lt;48 mmol/mol) after at least 2 months off all glucose-lowering medication</p> | <p>i) Number of participants achieving weight loss of 15kg or more at 6 months</p> <p>ii) Assessments of medication use and measurement of HbA1c concentration at 6 months</p> |
| <p><b>Secondary Objectives</b></p> <p>To compare effect of the Habitual Remission Programme versus standard usual care on:</p>                                                                               |                                                                                                                                                                                                |                                                                                                                                                                                |
| Glycaemic control                                                                                                                                                                                            | HbA1c                                                                                                                                                                                          | HbA1c at baseline, 3 and 6 months                                                                                                                                              |
| Weight                                                                                                                                                                                                       | Weight                                                                                                                                                                                         | Weight at baseline, 3 and 6 months                                                                                                                                             |
| Waist circumference                                                                                                                                                                                          | Waist circumference                                                                                                                                                                            | Waist circumference at baseline, 3 and 6 months                                                                                                                                |
| Systolic blood pressure                                                                                                                                                                                      | Systolic blood pressure                                                                                                                                                                        | Systolic blood pressure at baseline, 3 and 6 months                                                                                                                            |

|                                                                                      |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                         |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diastolic blood pressure                                                             | Diastolic blood pressure                                                                                                                                                                                                                                                                         | Diastolic blood pressure at baseline, 3 and 6 months                                                                                                                    |
| Medication use                                                                       | Medication use                                                                                                                                                                                                                                                                                   | Medication use at baseline, 3 and 6 months                                                                                                                              |
| i) Weight loss of at least 10kg in T2D patients and<br>ii) Remission in T2D patients | Number of participants with:<br>i) Weight loss of 10kg or more or<br>ii) HbA1c of less than 6.5% (<48 mmol/mol) after at least 2 months off all glucose-lowering medication                                                                                                                      | i) Number of participants achieving weight loss of 10kg or more at 6 months<br><br>ii) Assessments of medication use and measurement of HbA1c concentration at 6 months |
| Safety                                                                               | Evaluation of overall safety of Habitual Remission Programme by the monitoring of: <ul style="list-style-type: none"> <li>• Number of AEs</li> <li>• Number of SAEs</li> <li>• Number of (S)AEs that constitute Major Adverse Cardiovascular Events and Major Adverse Diabetes Events</li> </ul> | For the 6 Month duration of the trial                                                                                                                                   |
| Intervention Adherence                                                               | Number of participants completing the intervention at 3 and 6 months                                                                                                                                                                                                                             | Number of participants starting the intervention at baseline compared to those completing the intervention at 3 and 6 months                                            |

#### 4.2 Observational Study (12-month follow-up in the intervention group only)

| Objectives                                                                                                                                         | Outcome Measures | Timepoint(s) of evaluation of this outcome measure |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------|
| <b>Secondary Objectives</b><br>To assess whether at 12 months, diet replacement and a digital therapeutic (Habitual Remission Programme) leads to: |                  |                                                    |

|                                                                                         |                                                                                                                                                                                |                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| i) Weight loss of at least 15kg in T2D patients.<br>or<br>ii) Remission in T2D patients | Number of participants with:<br>i) Weight loss of 15kg or more<br>or<br>ii) HbA1c of less than 6.5% (<48 mmol/mol) after at least 2 months off all glucose-lowering medication | i) Number of participants achieving weight loss of 15kg or more at 12 months.<br><br>ii) Assessments of medication use and measurement of HbA1c concentration at 12 months  |
| i) Weight loss of at least 10kg in T2D patients.<br>or<br>ii) Remission in T2D patients | i) Weight loss of 10kg or more<br>or<br>ii) HbA1c of less than 6.5% (<48 mmol/mol) after at least 2 months off all glucose-lowering medication                                 | i) Number of participants achieving weight loss of 10kg or more at 12 months.<br><br>ii) Assessments of medication use and measurement of HbA1c concentration at 12 months. |

## 5 STUDY DESIGN

### 5.1 Study Type

Prospective, open-label, randomised controlled, parallel design with an active control.

### 5.2 Duration

Participants will be assigned to the intervention or standard usual care for 6 months, with an additional 1-week onboarding prior to this. In addition, HbA1c results will be collected at 12 months, via home blood test samples analysed by a central laboratory or from GP data, as an observational study (see section 7.11).

## 6 PARTICIPANT IDENTIFICATION

### 6.1 Study population

Participants with T2D diagnosed within the previous 6 years not on insulin (see full inclusion/exclusion criteria below).

#### 6.1.1 Inclusion Criteria

Participants must meet all the following criteria:

1. Able and willing to give consent for the study prior to participation
2. Be aged 18 - 75 years, with type 2 diabetes mellitus of duration <6 years
3. Have access to a smartphone or a computer
4. Have a Body Mass Index (BMI) of at least 28 kg/m<sup>2</sup>
5. A HbA1c test >48 mmol/mol (6.5%) and ≤86 mmol/mol (10%), within the previous 12 months



### 6.1.2 Exclusion criteria

Participants meeting **any** of the following criteria are not eligible for inclusion in the study :

1. Are currently using insulin
2. Weight change of >5% in the past 3 months
3. Have a history or are known to be suffering with alcohol/substance abuse
4. Have cancer or currently under investigation for cancer
5. Have had a myocardial infarction within the previous 6 months
6. Have severe or unstable heart failure e.g. NYHA grade IV
7. Have porphyria
8. Have learning difficulties
9. Are currently on treatment with anti-obesity drugs
10. Have had bariatric surgery
11. Have been diagnosed with eating disorder or purging
12. Are pregnant or less than 4 months postpartum or considering pregnancy in the next 2 years
13. Are currently breastfeeding
14. Have required hospitalisation for depression or taking antipsychotic drug
15. Have a history of illnesses that could interfere with the interpretation of the study results (e.g. HIV, Cushing syndrome, chronic kidney disease, chronic liver disease, hyperthyroidism, hereditary fructose intolerance, depression or antipsychotic drug use within the past 2 years)
16. Currently taking Glucagon-like peptide-1 receptor agonists (GLP-1 RAs)
17. Have pancreatitis
18. Currently taking part in a CTIMP trial for antidiabetic medication
19. Abnormal diabetic foot review (QOF codes for diabetic foot at moderate risk, at high risk, at increased risk, ulcerated).

## 6.2 Recruitment of Participants

Participants will be identified through GP practices within the Lindus Health Primary Care Network and referred to the central Lindus Health research team, to recruit the participant using the procedures outlined below.

To facilitate recruitment as required and depending on the success of recruitment through primary care, the following strategies may be implemented:

1. Social media advertisements
2. Engagement with the Lindus Health patient community

## 7 TRIAL PROCEDURES

Trial procedures are detailed in the Trial Flow Chart (Appendix 1). Activities requiring the involvement of the participant's GP are highlighted in orange.

### 7.1 Digital Applications

For the duration of the trial participants will be asked to record their measurements directly into the Lindus Health EDC platform, via a participant survey link, which will be emailed to them. The participants in the intervention arm will also use the Habitual app in parallel. Certain fields may require duplicate data entry between the two digital systems, but it is essential that data on the Lindus Health platform is completed fully.

The Lindus Health team will have access to the Lindus Health data and will review this on an ongoing basis. Any missing data will trigger an alert message to the participant via text/email. If no response is entered, a daily reminder will be triggered for 3 days, after which point the Lindus Health team will contact the participant to assess if they wish to withdraw from the study. If a participant wishes to withdraw then they will be advised to follow-up with their GP and a reason for withdrawal will be documented. Participants who wish to withdraw from completing home blood tests may continue in the trial and carry on recording their weight (primary outcome) and secondary outcome data in the participant survey.

The Sponsor (Habitual), and if required, the Chief Investigator will review the data from the Habitual App. If there are any causes for concern, the Sponsor will inform the Lindus Health team and the participant's GP will be contacted.

## **7.2 Pre-screening**

The participant will be directed to the trial web page, where they will be informed about the trial. If interested in taking part, they will complete an online pre-screening form to assess their eligibility for the trial. If potentially eligible and once they submit the form, the PIS and ICF will be emailed directly to the participant and they will be asked to book a telephone/video call with the Research Nurse (RN), during which they will complete informed consent and screening.

## **7.3 Informed consent**

Prior to consent, participants will have received the Patient Information Sheet (PIS), and Informed Consent Form (ICF) via email and had the opportunity to read through the details relating to the exact nature of the trial, the known side-effects and any risks involved in taking part.

Participants will be asked to provide informed consent only after a discussion with the RN and prior to randomisation. Adequate time will be given to the participant to consider the information and to ask any questions about the trial before deciding whether to participate. If the participant is willing, evidence of informed consent will be obtained. The participant will provide their electronic signature at the end of the e-consent form and the RN will countersign the form. A copy of the fully executed ICF will be sent to the participant and a copy retained by Lindus Health in a secure area with restricted access.

Following consent, the participant will be registered onto the Lindus Health system (confidential details, including name and date of birth, will be entered onto a secure area of the Lindus Health database, with restricted access).

## **7.4 Screening and Eligibility Confirmation**

Screening will be conducted remotely by the RN, during the same telephone/video call where informed consent is obtained. The RN will use the information provided by Habitual together with

medical history obtained from the participant, to confirm that the participant meets the inclusion and exclusion criteria. A record of screen failures that do not meet the inclusion/exclusion criteria will be retained.

GPs will then be notified and asked to contact us within 7 days (via email / phone) if they have any objections to the inclusion of the patient in the trial.

### **7.5 Randomisation**

Following confirmation of eligibility, the participant will be randomised using a secure, fully validated and compliant web-based randomisation system called Sealed Envelope.

Participants will be randomised into one of the two groups at a ratio of 2:1, stratified by gender.

1. **Control arm:** participants receive standard usual care
2. **Intervention arm:** participants will receive the Habitual Remission Programme

The RN will call the participant to inform them of their randomisation allocation and they will also be sent an email confirmation.

### **7.6 Participant GP review appointment - Intervention Group only**

Participants randomised to the intervention arm will be asked to schedule a telephone appointment with their GP to discuss the study, and their withdrawal from all glucose-lowering medication prior to starting TDR. The appointment will occur during the participant onboarding period (see below), and the intervention will only begin after the GP has reviewed the participants' inclusion in the trial.

### **7.7 Participant Onboarding**

Following randomisation and prior to baseline, participants will undergo an onboarding period to assess if they would like to take part in the trial, and to allow time for the GP review visit for those in the intervention group. During the onboarding period participants in the intervention group will continue to use their glucose-lowering medication as normal.

The participant will be issued with measuring equipment: blood pressure monitor, tape measure and weighing scales and asked to record blood pressure, weight, waist circumference directly into the EDC platform via a survey link. During the onboarding period, participants will indicate in the survey when they are ready to start the trial stage. For participants in the intervention arm, they will need to confirm that the GP review visit has taken place, at which point the Habitual TDR (including instructions) will be issued.

### **7.8 Baseline Measurements (week 1, day 1 to week 2)**

Once continuation has been confirmed and prior to starting the intervention, participants will collect baseline measurements and record these in their survey:

- Measure blood pressure, weight, waist circumference
- Collect the HbA1c blood sample using the kit provided and send for central processing in the pre-paid envelope.

## **7.9 Follow-up**

### **7.9.1 Intervention Group - Intensive Blood Pressure (BP) Monitoring**

Participants in the intervention group will be asked to measure their blood pressure once a week for the first two weeks after starting TDR. These BP readings will be captured in the participant survey, and this data will be reviewed on a regular basis by the central research team, who will notify the GP of any results which require follow-up. Please see Appendix 3 for details of BP monitoring, medication withdrawal and reintroduction. All participants in the intervention arm will also receive a phone call from the RN, 4 weeks after starting TDR, to identify any potential issues relating to their blood pressure recordings.

### **7.9.2 Fortnightly checks: (weeks 2, 4, 6, 8, 10, 14, 16, 18, 20, 22)**

Participants will be asked to record the following in their survey:

- AEs: any potential side-effects
- SAEs such as hospitalisation (for 24 hours or more), prolongation of hospitalisation or any debilitating effects
- Any change in medication
- Blood pressure, weight, waist circumference

### **7.9.3 Month 3 (Week 12)**

During week 12, participants will be asked to record all of the information described for the fortnightly checks (section 7.9.2), in addition to providing a blood sample using the kit provided, for HbA1c analysis.

Participants in the intervention arm will undergo food reintroduction via the Habitual app at this stage (please see section 8.1 for details).

### **7.9.4 Month 6 (Week 24) and End of Study**

During week 24, participants will be asked to record all of the information described for the fortnightly checks (section 7.9.2), in addition to providing a blood sample using the kit provided, for HbA1c analysis.

### **7.9.5 Participant Study Experience**

At the end of the study (6 Months) participants will be asked about their experience in the study, use of the digital technology and aspects that they may have found useful. The response data may be used for potential publication/media release in journals, newspapers, radio, interviews, TV, social media, websites, newsletters, marketing in print and online. Participant response data for such purposes will be anonymised (without participant's name or other personal identifiers). When being asked to provide responses, participants will be provided with a list of the above media types so that they can provide specific permission for which types of publication can be used.

### **7.10 Observational Study**

The randomised controlled trial (RCT) will end at 6 months after baseline. The intervention group only will undergo a further 6 months of follow-up to assess long-term remission. At 9 and 12 months, they will be asked to complete a survey on the EDC platform or information may be obtained from participant medical records, for which the participant will have provided consent. At 12 months, participants will be asked to provide a blood sample using the kit provided (for HbA1c analysis).

Those in the control group will be offered the Habitual Remission Programme at the end of the 6-month RCT, under the supervision of Habitual and not part of the observational study for the intervention group.

### **7.11 Early Discontinuation/Withdrawal of Participants**

Each participant has the right to withdraw from the study at any time. In addition, the GP or Sponsor may discontinue the participation of a participant from the study at any time, if this is considered necessary for any reason including:

- Pregnancy
- Ineligibility
- Significant protocol deviation
- Withdrawal of consent

Following participant withdrawal, no further clinical assessments will be performed. The participant will be contacted by a member of the study team and a reason for withdrawal will be requested and documented. All data recorded prior to withdrawal will be analysed in the intention-to-treat population.

### **7.12 Definition of End of Trial**

The end of the study is defined as the database lock date (ie. following completion of any follow-up monitoring and data collection, as described in the protocol).

## **8. STUDY INTERVENTION**

### **8.1 Intervention Group (intervention description)**

The Habitual Remission Programme combines a digital therapeutic with TDR.

#### ***Digital therapeutic***

The digital therapeutic uses a behaviour change programme, which will include psychological, emotional, and social support delivered via a digital therapeutic App. Participants will be taken through clinically-proven psychological techniques which will help to change the way they think about nutrition, mood, and motivation.

#### ***TDR***

The dietary weight loss product utilises formula foods, designed to replace an individual's entire diet for an extended period to facilitate significant weight loss. These formula foods are strictly

regulated (EU legislation - Regulation on Foods for Specific Groups 609/2013/EU) and provide guaranteed amounts of required nutrition (14). The diet is designed to achieve at least 10kg of weight loss.

The intervention will be administered as follows:

**1. Weeks 0 to 12 (month 0 - 3) - (TDR):**

The first three months of Habitual involves a diet of nutritionally complete TDR shakes and soups. Participants will have four 'meals' per day for a total 800kcal per day.

**2. Weeks 12 - 24 (month 3 - 6) - Food Reintroduction & calorie maintenance:**

In the second phase of the programme participants gradually reintroduce food into their diet, supported by the Habitual digital app. Food reintroduction will be managed over 3 months, with a 6 week period of increase in caloric intake, followed by 6 weeks of further nutritional education and management of caloric intake. At week 12, participants will consume one meal per day (shakes/soups for the rest of meals), at weeks 14-16 two meals, and weeks 16-18 three meals. Weeks 19-24 of the programme will help participants to find and maintain their 'new normal'.

## **8.2 Concomitant Therapy**

### **8.2.1 Medication withdrawal and reintroduction**

For participants in the intervention arm all glucose-lowering medications should be stopped in line with current recommended best practice and NICE Guidelines, on the same day that TDR is started. Following medication withdrawal and starting TDR, participants BP recordings will be monitored by the RN and Medical Lead, who will alert the participants' GP to any recordings which require further follow-up under standard usual care (see section 10.4.1). Diabetes medication may need to be reintroduced if T2D returns and antihypertensives reduced if hypotension occurs, using the NICE guidelines for medication withdrawal and reintroduction as summarised in Appendix 3.

Participants will also monitor their BP recordings and be provided with a guide informing them when to seek care from their GP (see section 8.2.3).

### **8.2.2 Prohibited Medications**

Total diet replacement or reduced calorie diets may affect other medications the participant has been prescribed. Routine medical review of participants by the GP will continue. Any medication detailed in the inclusion/exclusion criteria will be prohibited during the study.

### **8.2.3 BP Monitoring**

Participants in the intervention group will be advised to record their BP two days per week for the first 2 weeks and then fortnightly to allow careful monitoring of any clinically relevant changes in BP. Participants/Lindus Health team will contact the patient's GP using a traffic light system.

### 8.3 Control Group

Those allocated to the control arm of the study will continue standard usual care.

Usual Care refers to standard NHS care for patients with Type 2 diabetes (<https://www.nhs.uk/conditions/type-2-diabetes/going-regular-check-ups/>).

This involves checks at 3-6 and 12 months, delivered by the GP or diabetes nurse. Blood sugar levels (HbA1C test) are checked every 3 months when newly diagnosed, then every 6 months once stable. Once a year, feet (for ulcers and infections), eyes (for damage to blood vessels), and blood pressure, cholesterol and kidneys (for high blood pressure, heart and kidney disease) are reviewed.

In order to facilitate retention, participants in the control group will be offered the Habitual Remission Programme at the end of the 6-month RCT period.

### 8.4 Adherence

The number of participants continuing with the intervention at 3 and 6 months will be assessed, as a measure of adherence to the Habitual Remission Programme.

## 9. SAMPLE HANDLING

Participants will collect blood samples at baseline, 3, 6 and 12 months (as outlined in the schedule of events, Appendix 2), using a home blood test kit. They will be provided with instructions and a video link to ensure accurate administration of the test, and will be asked to send their sample in a pre-paid envelope for central HbA1c laboratory analysis.

## 10. SAFETY REPORTING

### 10.1 Definitions

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adverse Event (AE)    | Any untoward medical occurrence in a patient or clinical study participant.                                                                                                                                                                                                                                                                                                                                                                                              |
| Adverse Reaction (AR) | <p>An untoward and unintended response in a participant to an intervention.</p> <p>The phrase "response to an intervention" means that a causal relationship between an intervention and an AE is at least a reasonable possibility, i.e. the relationship cannot be ruled out. All cases judged by either the reporting medically qualified professional as having a reasonable suspected causal relationship to the trial medication qualify as adverse reactions.</p> |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serious Adverse Event (SAE)                | <p>A serious adverse event is any untoward medical occurrence that:</p> <ul style="list-style-type: none"> <li>• results in death</li> <li>• is life-threatening</li> <li>• requires inpatient hospitalisation or prolongation of existing hospitalisation</li> <li>• results in persistent or significant disability/incapacity</li> <li>• consist of a congenital anomaly or birth defect</li> </ul> <p>Other 'important medical events' may also be considered serious if they jeopardise the participant or require an intervention to prevent one of the above consequences.</p> <p>NOTE: The term "life-threatening" in the definition of "serious" refers to an event in which the participant was at risk of death at the time of the event; it does not refer to an event, which hypothetically might have caused death if it were more severe.</p> |
| Serious Adverse Reaction (SAR)             | An adverse event that is both serious and, in the opinion of the reporting Investigator, believed with reasonable probability to be due to one (or lack of) of the intervention, based on the information provided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Related and unexpected SAEs                | <p>All study SAEs in the opinion of the Chief Investigator or designated person, to be:</p> <ul style="list-style-type: none"> <li>• 'related', i.e. resulted from the administration of any of the research procedures; and</li> <li>• 'unexpected', i.e. an event that is not listed in the protocol as an expected occurrence</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Major Adverse Cardiovascular Events (MACE) | Acute myocardial infarction (AMI), stroke, cardiovascular death, unstable angina and heart failure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Major Adverse Diabetes Events (MADE)       | Eye problems (retinopathy), kidney problems (nephropathy), nerve damage (neuropathy), other related conditions, like cancer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## 10.2 Causality

The relationship of each adverse event to the intervention must be determined by a medically qualified individual according to the following definitions:

- **Unrelated** – where an event is not considered to be related to the intervention
- **Possibly** – although a relationship to the intervention cannot be completely ruled out, the nature of the event, the underlying disease, concomitant medication or temporal relationship make other explanations possible
- **Probably** – the temporal relationship and absence of a more likely explanation suggest the event could be related to the intervention.
- **Definitely** – the known effects of the intervention, its therapeutics class or based on challenge testing suggest that the intervention is the most likely cause.



All AEs (SAEs) labelled possibly, probably or definitely will be considered as related to the intervention.

### **10.3 Adverse Event Reporting**

(S)AEs will be captured for the 6-month trial duration from all participants (those in the intervention and control arms). They will be asked in their fortnightly surveys if they have experienced any side effects or medical events while participating in the DIGEST trial. Participants may also report potential (S)AEs via telephone calls or email correspondence with the Research Nurse.

To assess the impact of the intervention on cardiovascular health, and diabetes complications, the proportion of (S)AEs which constitute Major Adverse Cardiovascular Events (MACE) and Major Adverse Diabetes (MADE) will be recorded.

#### **10.3.1 Known side-effects**

The following side-effects are known to be associated with TDR use and the participant will be made aware of such events in the PIS.

- Halitosis (bad breath).
- Abdominal pain.
- Diarrhoea.
- Headache.
- Hair loss.
- Dry skin.
- Mood changes.
- Chills/ feeling cold.
- Tiredness/fatigue.

### **10.4 Measures to minimise the occurrence of AEs**

#### **10.4.1 Blood Pressure Monitoring in the Intervention Group**

Participants will self-monitor their blood pressure once a week for the first two weeks after starting TDR. They will then measure their BP on a fortnightly basis. Throughout the trial they will be asked to use the traffic light system (see Appendix 4) to ensure that they seek medical care as required. To further prevent any potential safety events, the central Lindus Health team will also monitor BP readings entered into the participant survey and using the traffic light system (see Appendix 4), alert the participant and their GP to any BP measurements which require follow-up. All participants in the intervention arm will also receive a phone call from the RN, 4 weeks after starting TDR, to answer any queries and to specifically identify any potential issues relating to BP measurements.

### **10.5 SAE Reporting**

Any AEs which fall under the definition of a SAE will be reported to the Chief Investigator or designated person within 24 hours, to be reviewed for relatedness and expectedness. However, relapse and hospitalisations for elective treatment of a pre-existing condition do not need reporting as SAEs.

Reports of related and unexpected SAEs (in the opinion of the Chief Investigator or designated person) will be reported to the REC within 15 days of the CI and/or study team becoming aware of the event, using the REC SAE form for non-CTIMP studies. The Chief Investigator will also notify the Sponsor of all study SAEs.

## **10.6 Pregnancy**

Prior to randomisation, female participants of child-bearing potential will be asked to confirm that they are not pregnant or less than 4 months postpartum, and not considering pregnancy for the next 2 years. During follow-up they will be asked whether they have become pregnant, in the fortnightly survey. Any reports of pregnancy will trigger a safety alert to the trial team. In addition, if a female participant becomes pregnant during the study, they will be informed in the PIS that this should be reported immediately to the Lindus Health team.

Any pregnancy occurring during the study will result in the immediate discontinuation of the participant and the Sponsor should be notified within 24 hours so appropriate follow-up is conducted. Pregnancy itself is not an AE.

## **11. STATISTICS**

The Sponsor/designee will be responsible for the statistical analysis, in accordance with the Statistical Analysis Plan.

### **11.1 Description of Statistical Methods**

The primary analysis will be by intention-to-treat (ITT). Participants reporting diet intolerance or poor compliance will be recorded and these participants will be included in the ITT analysis.

Primary outcome measures will be analysed in a hierarchical fashion, first analysing reduction in weight of 15 kg or more as a binary outcome with significance assessed at the 5% significance level, followed, if the first test is significant, by test of remission of diabetes status as a binary outcome also at the 5% level of significance. The overall type I error will not exceed 5% because of the hierarchical nature of the testing. For participants who discontinue the formal weight management programme the follow-up weights (from routine attendances) and the end of study diabetes status will be used.

As a secondary analysis of the primary outcome, a causal effects analysis will be completed to estimate the effect of the intervention in the subset of people who comply with the intervention. Participants who completed the intervention will be defined as a participant in the intervention arm who completed the Habitual Remission Programme at 6 months, irrespective of their dietary adherence or weight-change. Fisher's Exact Test will be used for the primary outcome.

The difference between the intervention and usual care in the following secondary outcomes will be analysed under the ITT principle: HbA1c, weight, waist circumference, blood pressure, medication use, T2D remission and weight loss of 10 kg or more, safety, and intervention adherence, at the different time points (Baseline and Month 6). The observational study will compare T2D remission and at least 15kg of weight loss, and T2D remission and at least 10kg of weight loss, at 12 months. Outcomes will be compared between groups with mixed-effects

regression models. Logistic models used for binary outcomes, and Gaussian models for continuous outcomes.

### **11.2 The Number of Participants**

Sample size calculations are based on the remission rates of DiRECT at 12 months (46% in the intervention group and 4% in the control). Conservative calculations have assumed diabetes remission in 32% of intervention participants at 6 months compared with 4% in the control group. Allowing for an estimated 20% individual participant drop-out within 6 months, we will recruit a total of 100 participants to provide 90% power to detect significant differences. Participants will be recruited in cohorts using a 2:1 ratio (intervention:control).

### **11.3 Criteria for termination of the trial**

It is not anticipated that the trial will be suspended or terminated unless in the event of safety concerns or futility such as poor recruitment rates. The study may resume once concerns are addressed and with agreement from the CI, Sponsor and Ethics committee.

### **11.4 Handling of Missing and Incomplete Data**

Missing data will be reported with reasons given where available. Depending on the extent of missing values, further investigation may be made into the sensitivity of the analysis results to the method(s) specified. The primary analysis will assume missing outcomes are failures.

## **12 DATA MANAGEMENT**

### **12.1 Source Data**

Source documents are where data are first recorded, and from which participants' survey data are obtained. These include, but are not limited to, medical records, pharmacy records, participant surveys, and correspondence.

EDC entries will be considered source data if the EDC is the site of the original recording (e.g. there is no other written or electronic record of data; e.g. survey blood pressure measurements). All documents will be stored securely and safely. In the EDC system, the participant will be referred to by the trial participant number, not by name.

### **12.2 Access to Data**

Data will be entered into a validated EDC platform. Direct access to the EDC platform will be granted to authorised representatives from the Sponsor, Lindus Health and the regulatory authorities to allow trial-related monitoring, audits and inspections. To ensure data transparency, the trial will be registered on a publicly available database for clinical studies.

### **12.3 Data Handling and Record Keeping**

Informed consent (e-consent) and eligibility criteria will be completed using worksheet templates, which will be retained in restricted access secure area. The remaining data will be collected directly onto the EDC platform, therefore no further source will be available. Central Laboratory data will be transferred electronically to Lindus Health study database under a laboratory transfer agreement.

An online secure data entry system designed to collect sensitive data, such as participant and Study Partner contact details, will be used. All identifiable participant data is encrypted using the Advanced Encryption Standard. The participant portal will also manage online ePRO. Participant data will be kept and stored securely for as long as it's required by the study and reviewed on an annual basis.

### **13. QUALITY ASSURANCE PROCEDURES**

The study may be monitored, or audited in accordance with the current approved protocol, GCP, relevant regulations and standard operating procedures.

#### **13.1 Risk Assessment**

A risk assessment and categorisation tool will be prepared before recruitment starts and will be reviewed and amended as appropriate during the study, to reflect significant changes to the protocol or to the study conduct.

#### **13.2 Study Monitoring**

Monitors may verify that the clinical study is conducted, and data are generated, documented and reported in compliance with the protocol, GCP and the applicable regulatory requirements. Specifics will be detailed in the Trial Monitoring Plan.

#### **13.3 Study Committees**

The Lindus Health Clinical Operations Team, Medical Lead and Product Manager will work closely with the CI and Sponsor to ensure compliant trial conduct. Frequent meetings will be held to discuss trial progress and to review and resolve any potential issues.

### **14. PROTOCOL DEVIATIONS**

A study related deviation is a departure from the ethically approved study protocol or other study document or process (e.g. consent process or administration of study intervention) or from Good Clinical Practice (GCP) or any applicable regulatory requirements. Any deviations from the protocol will be documented in a protocol deviation form and filed in the electronic Trial Master File (eTMF), in line with Lindus Health SOPs.

### **15. SERIOUS BREACHES**

A “serious breach” is a breach of the protocol or of the conditions or principles of Good Clinical Practice which is likely to affect to a significant degree

- (a) the safety or physical or mental integrity of the trial subjects; or
- (b) the scientific value of the research.

In the event that a serious breach is suspected the Sponsor must be contacted within one working day. In collaboration with the CI, the serious breach will be reviewed by the Sponsor and, if appropriate, the Sponsor will report it to the approving REC committee and the relevant NHS host organisation within seven calendar days.

## **16. ETHICAL AND REGULATORY CONSIDERATIONS**

### **16.1 Declaration of Helsinki**

The Investigator will ensure that this study is conducted in accordance with the principles of the Declaration of Helsinki.

### **16.2 Guidelines for Good Clinical Practice**

The Investigator will ensure that this trial is conducted in accordance with relevant regulations and with Good Clinical Practice.

### **16.3 Approvals**

The protocol, informed consent form, participant information sheet and any proposed advertising material will be submitted to an appropriate Research Ethics Committee (REC) and HRA for written approval. The Investigator will submit and, where necessary, obtain approval from the above parties for all substantial amendments to the original approved documents.

### **16.4 Reporting**

An Annual Progress Report will be submitted once a year to the REC, host organisation and Sponsor. In addition, an End of Trial notification and final report will be submitted to the REC, host organisation and Sponsor.

### **16.5 Participant Confidentiality**

The trial staff will ensure that the participants' anonymity is maintained. The participants will be identified only by a participant ID number on all trial documents and in the electronic clinical database. All data will be stored securely on an electronic study database and will comply with the General Data Protection Regulation (GDPR) and Data Protection Act 2018, which require data to be de-identified as soon as it is practical to do so. The study database will be managed according to Standard Operating Procedures maintained by Lindus Health. Access rights to data and applications software will be clearly defined and staff authorised to access personal data will be formally notified in writing of the permissible scope of their access. Data access will be limited to specific members of the research team (trained in data protection policy) including the chief investigator (as study guarantor), and programmers. For each database application, system users will be given a valid user system account name (username ID), and a password known only to that user to prevent unauthorised use of systems. All data will be entered into the database through a reliably encrypted gateway.

### **16.6 Expenses and Benefits**

Patients in the intervention arm of the study will be provided with clinically validated BP monitoring equipment during the trial. Those in the control group will receive a free subscription to the Habitual Remission Programme at the end of the 6 month trial period.

### **16.7 Patient and Public Involvement (PPI)**

The Lindus Health Patient Advisory Board has provided substantial input to the protocol and the overall study design. The EDC platform has undergone user testing to ensure that it is user friendly and the language is understood.

## **17. FINANCE AND INSURANCE**

### **17.1 Funding**

This study will be funded by Habitual and conducted by Lindus Health. Blood samples will be collected by the participant at home and processed centrally. All laboratory kits and courier costs will be paid for by the Sponsor.

### **17.2 Insurance**

The Sponsor has undertaken insurance for this study.

## **18. PUBLICATION POLICY**

The Chief Investigator will be involved in reviewing drafts of the manuscripts, abstracts, press releases and any other publications arising from the study. Authors will acknowledge that the study was funded via a clinical trial agreement between Habitual and Lindus Health. Authorship will be determined in accordance with the International Committee Medical Journal Editors guidelines and other contributors will be acknowledged.

No publication or disclosure of study results will be permitted except under the terms and conditions of a separate written agreement between Habitual and the CI.

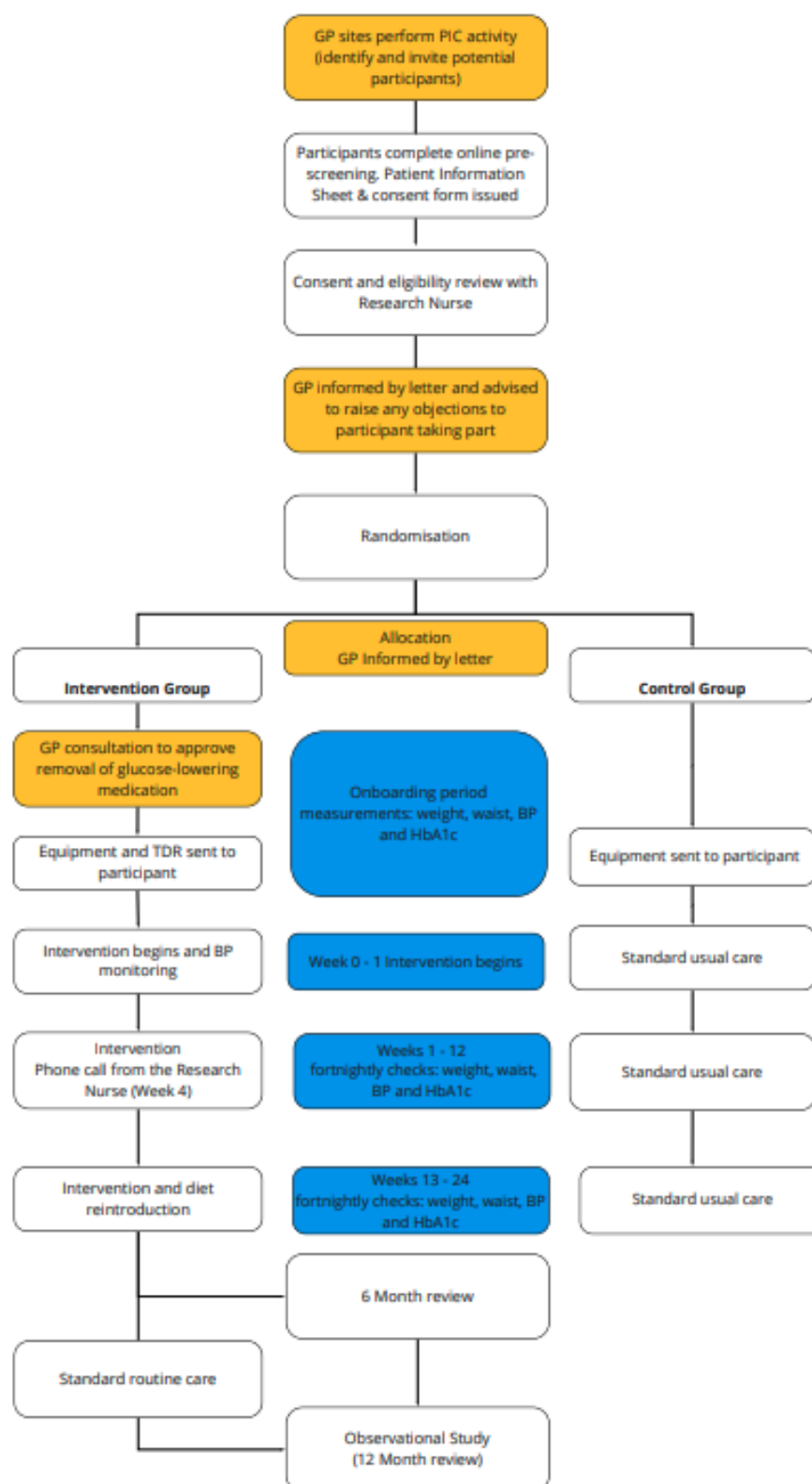
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## 20 APPENDIX 1 – TRIAL FLOW-CHART



## 21 APPENDIX 2 - SCHEDULE OF EVENTS TABLE

| Schedule of events                                                                               | Screening | Participant onboarding period | Baseline (Week 0) | Week 1 | Weeks 2, 4, 6, 8, 10 | 3 month (Week 12) | Weeks 14, 16, 18, 20, 22 | 6 month and End of RCT Study (Week 24) | 9 month (observational study) | 12 month (observational study) |
|--------------------------------------------------------------------------------------------------|-----------|-------------------------------|-------------------|--------|----------------------|-------------------|--------------------------|----------------------------------------|-------------------------------|--------------------------------|
| Control and Intervention Groups                                                                  |           |                               |                   |        |                      |                   |                          |                                        |                               |                                |
| Informed e-consent and eligibility confirmation                                                  | X         |                               |                   |        |                      |                   |                          |                                        |                               |                                |
| Randomisation                                                                                    |           | X                             |                   |        |                      |                   |                          |                                        |                               |                                |
| Telephone/video call from Research Nurse to inform participant of randomisation group allocation |           | X                             |                   |        |                      |                   |                          |                                        |                               |                                |
| Confirmation of continuation into the Trial                                                      |           |                               | X                 |        |                      |                   |                          |                                        |                               |                                |
| (Serious) Adverse Event recording                                                                |           |                               |                   |        | X                    | X                 | X                        | X                                      |                               |                                |
| HbA1c (home test)                                                                                |           |                               | X                 |        |                      | X                 |                          | X                                      |                               |                                |
| Equipment issued                                                                                 | X         |                               |                   |        |                      |                   |                          |                                        |                               |                                |
| Blood pressure measurement                                                                       |           | X                             | X                 |        | X                    | X                 | X                        | X                                      |                               |                                |
| Weight measurement                                                                               |           | X                             | X                 |        | X                    | X                 | X                        | X                                      |                               |                                |
| Waist circumference measurement                                                                  |           | X                             | X                 |        | X                    | X                 | X                        | X                                      |                               |                                |
| EDC Completion by Participant                                                                    |           |                               | X                 | X      | X                    | X                 | X                        | X                                      |                               |                                |

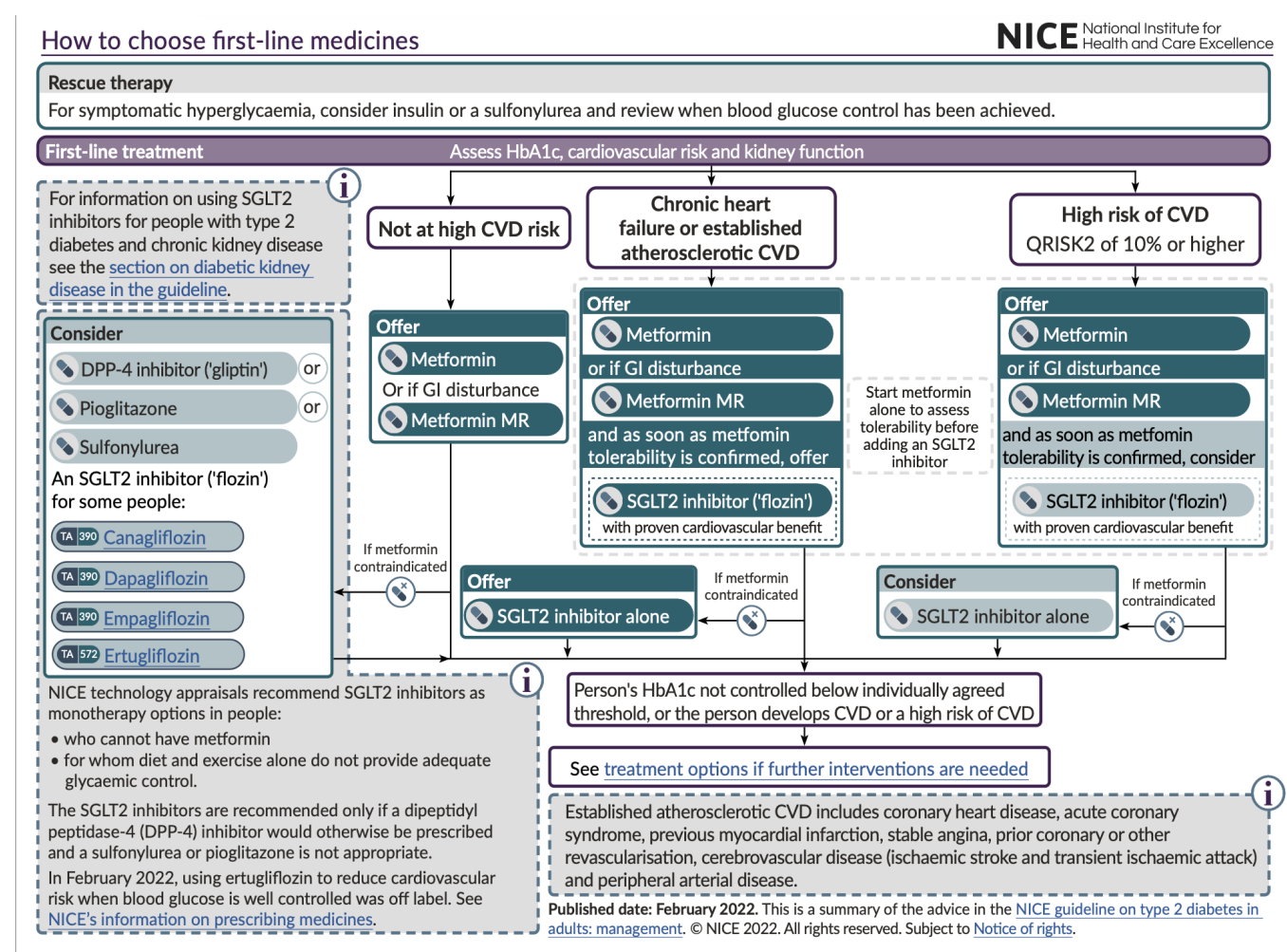
| Intervention Group Only                                |  |   |   |   |   |   |   |   |   |   |
|--------------------------------------------------------|--|---|---|---|---|---|---|---|---|---|
| GP appointment to discuss medication withdrawal        |  | X |   |   |   |   |   |   |   |   |
| Intervention started: Habitual app activated           |  |   | X |   |   |   |   |   |   |   |
| Total diet replacement (TDR) issued (monthly)          |  |   | X |   | X | X | X |   |   |   |
| Habitual App interaction                               |  | X | X |   | X | X | X | X |   |   |
| Intensive BP monitoring (weeks 1-2)                    |  |   | X | X | X |   |   |   |   |   |
| Telephone/video call from Research Nurse (week 4 only) |  |   |   |   | X |   |   |   |   |   |
| Additional HbA1c (home test)                           |  |   |   |   |   |   |   |   |   | X |
| Additional Blood pressure measurement                  |  |   |   |   |   |   |   |   | X | X |
| Additional Weight measurement                          |  |   |   |   |   |   |   |   | X | X |
| Additional Waist circumference measurement             |  |   |   |   |   |   |   |   | X | X |
| Additional EDC Completion by Participant               |  |   |   |   |   |   |   |   | X | X |

## 22 APPENDIX 3 – CURRENT NICE GUIDANCE FOR DIABETES AND BLOOD PRESSURE MEDICATION WITHDRAWAL AND MEDICATION REINTRODUCTION

### i) Diabetes Guidance

Please see NICE Guidelines for the management of Type 2 diabetes, <https://www.nice.org.uk/guidance/NG28> and antihypertensive drug treatment (<https://www.nice.org.uk/guidance/ng136/resources/visual-summary-pdf-6899919517>) in adults. Summary shown below.

### NICE Guidelines for guidance on the management of Type 2 diabetes



## 23 APPENDIX 4 – TRAFFIC LIGHT SYSTEM FOR MONITORING BLOOD PRESSURE MEASUREMENTS IN THE INTERVENTION GROUP

| Colour | Level  | Blood Pressure                                                  | Participant Action                                                                                                                                                                           |
|--------|--------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RED    | HIGH   | Systolic (SYS) 180 or more<br>OR<br>Diastolic (DIA) 120 or more | <b>Your BP is too high.</b><br><br>Make an appointment the same day to see your GP or nurse, or call NHS 111 .<br><br>If you have any new symptoms or are feeling unwell seek immediate help |
| AMBER  | RAISED | Systolic (SYS) 140-179<br>OR<br>Diastolic (DIA) 90-119          | <b>Your BP is raised.</b><br><br>If you have persistent AMBER readings (4 or more days of the week) then you should contact your GP/Practice nurse as you may need your medication altered.  |
| GREEN  | NORMAL | Systolic (SYS) 90-139<br>AND<br>Diastolic (DIA) 60-89           | <b>Your BP is normal.</b><br><br>This is fine provided that you have no side effects.                                                                                                        |
| BLUE   | LOW    | Systolic (SYS) 89 or less<br>OR<br>Diastolic (DIA) 59 or less   | <b>Your BP is too low.</b><br><br>Make an appointment the same day to see your GP or nurse or call NHS 111 . If you have any new symptoms or are feeling unwell seek immediate help          |

## 24 APPENDIX 5 – AMENDMENT HISTORY

| Amendment No.         | Protocol Version No. | Date issued   | Author(s) of changes | Details of Changes made                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------|----------------------|---------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (Pre REC approval)  | 1.1                  | 31-Aug-2022   | Hannah Swayze        | For clarity updates have been made to: wording regarding eligibility confirmation, Appendix 1 - Trial Flow Chart and Appendix 2 - Schedule of Events.                                                                                                                                                                                                                                      |
| 2 (Pre REC approval)  | 2.0                  | 18-Oct-2022   | Hannah Swayze        | Two additional eligibility criteria added (recommended by the REC), randomisation method updated to include stratification by gender, and Schedule of Events updated for clarity.                                                                                                                                                                                                          |
| 3 (post REC approval) | 2.1                  | 10- Jan- 2023 | Danni Maas           | Clarification of the time window for the participant onboarding period & associated amendments in the schedule of events and flow diagram. Minor typo amendments.                                                                                                                                                                                                                          |
| 4 (post REC approval) | 2.2                  | 19Sep2023     | <i>Danni Maas</i>    | <i>Amendment to the food reintroduction period for patients on the intervention arm, to clarify that food should be reintroduced over 6 weeks (instead of 12), followed by 6 weeks of further nutritional support &amp; management of caloric intake. Minor edits elsewhere in the protocol to ensure consistency is maintained whenever the food reintroduction process is mentioned.</i> |

Lists details of all protocol amendments whenever a new version of the protocol is produced. Protocol amendments must be submitted to the Sponsor for approval prior to submission to the REC committee and HRA (where required)